



Overview



Performance Analysis



3G Analysis



4G Analysis



5G Analysis



6G Prediction



Key Influencers



Conclusion



NEXT GENERATION MOBILE NETWORK ANALYSIS DASHBOARD(3G|4G|5G|6G PREDICTION)

Real-time insights into network, performance, signal quality, connectivity, and user experience

28.15

↑ Avg Speed

21.28

⌚ Avg Latency

-89.04

📶 Avg Signal Strength

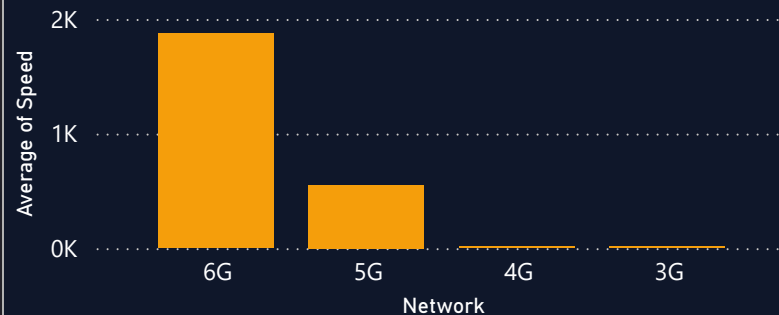
5.00

📶 Average of Range

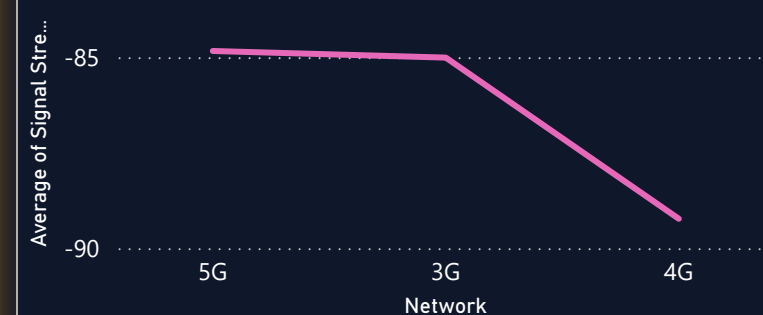
4

📶 Total Network

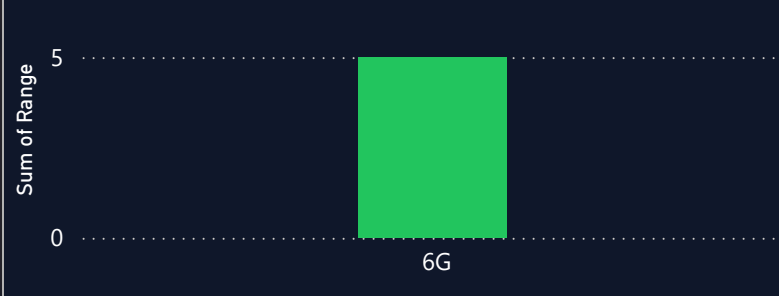
Average Speed by Network



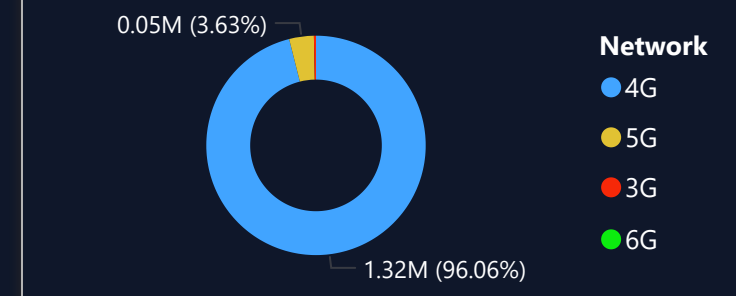
Average Signal Strength by Network



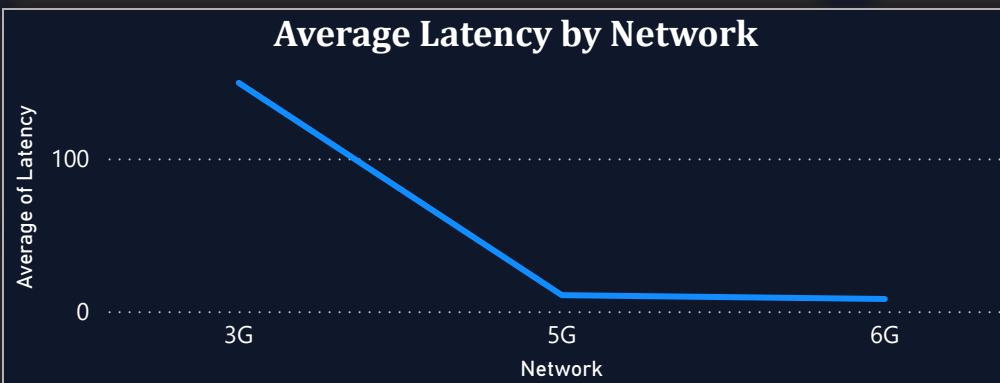
Average Range By Network



Network Distribution



Average Latency by Network



FILTERS



PERFORMANCE ANALYSIS

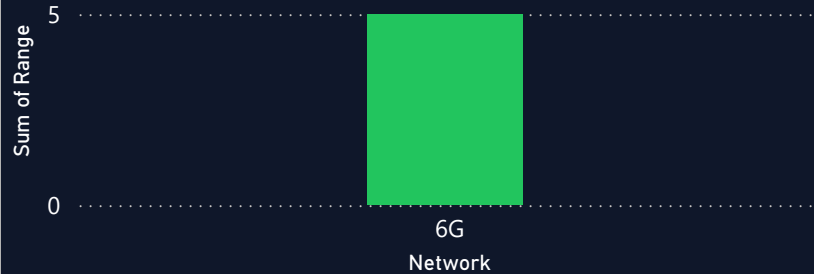
Analyze and compare network performance metrics across 3G,4G,5G and predicted 6G networks



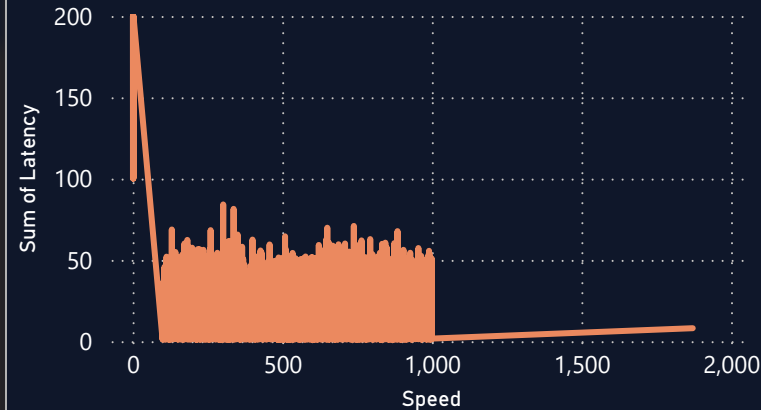
Network

All

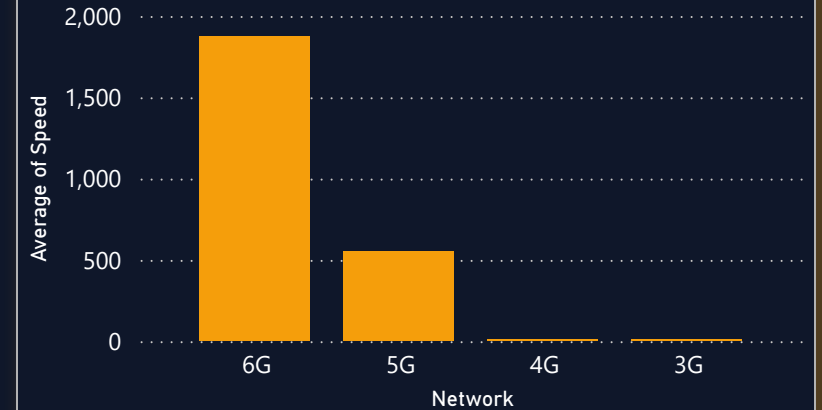
Average Range By Network



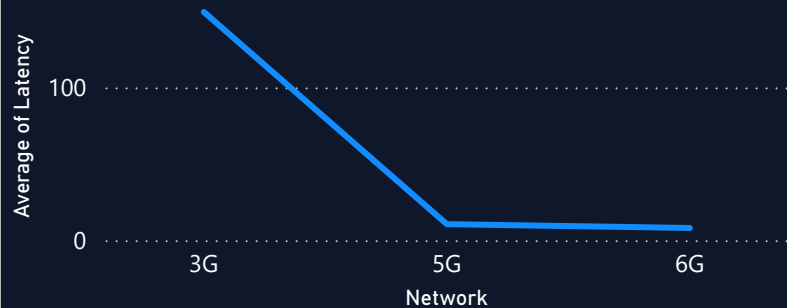
Speed Vs Latency



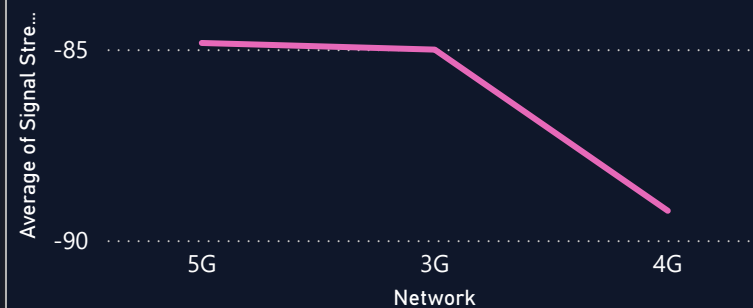
Average Speed by Network



Average Latency by Network



Average Signal Strength by Network



Network	Sum of Speed	Sum of Latency	Sum of Signal Strength
3G	8,442.77	6,28,736.71	-3,57,726.20
4G	1,12,03,409.93		-10,89,47,537.00
5G	2,75,59,053.69	5,24,862.20	-42,41,863.70
6G	1,872.19	8.00	
Total	3,87,72,778.58	11,53,606.90	-11,35,47,126.90

28.15

↑ Avg Speed

21.28

⌚ Avg Latency

-89.04

📶 Avg Signal Strength

5.00

📶 Average of Range

4

📶 Total Network

FILTERS

Timestamp

All

Locality

All

2.01

 Avg Throughput

149.41

 Avg Latency

-85.01

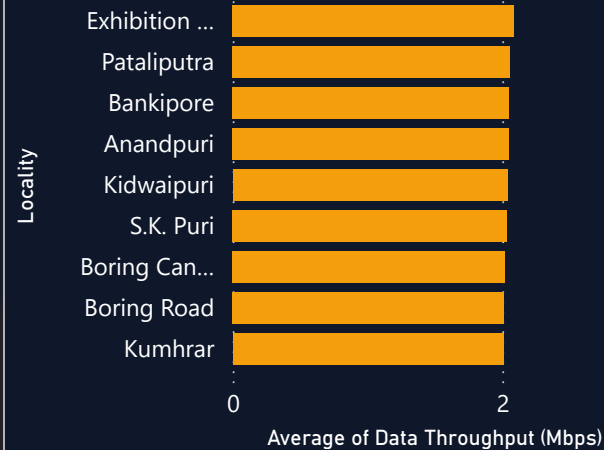
 Avg Signal Strength



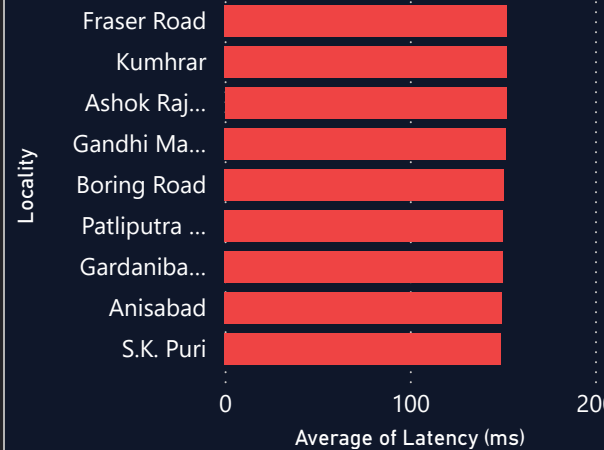
3G NETWORK ANALYSIS

Analyze the performance, latency, throughput, and signal quality of the 3G network

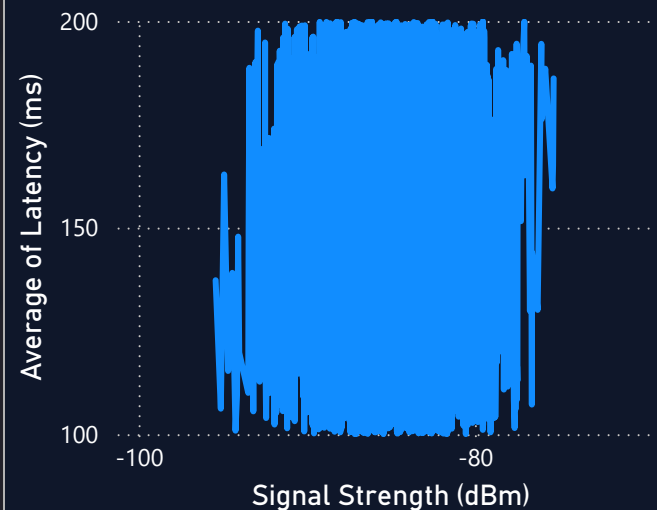
Throughput by Locality



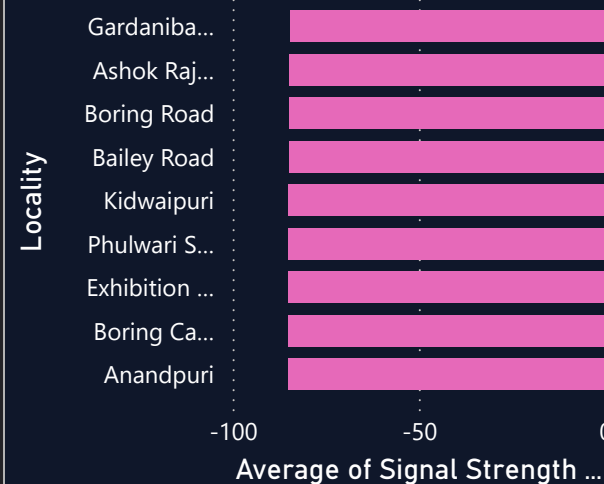
Latency by Locality



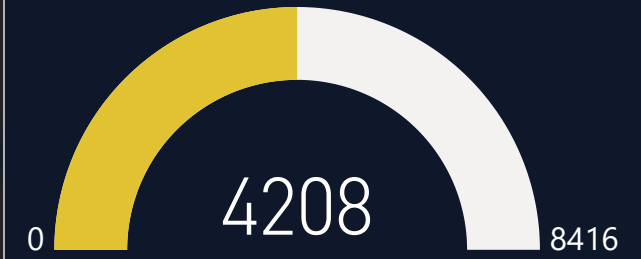
Signal Strength Vs Latency



Signal Strength by Locality



Total Network Type



Locality	Sum of Data Throughput (Mbps)
Kumhrar	472.3
Fraser Road	472.2
Bailey Road	449.4
Pataliputra	448.6
Ashok Rajpath	437.7
Gandhi Maidan	433.5
Phulwari Sharif	432.6
Kankarbagh	428.3
S.K. Puri	427.6
Anandpuri	426.3
Kidwaipuri	424.3
Anisabad	423.2
Bankipore	422.9
Boring Canal Road	416.6
Gardanibagh	411.9
Exhibition Road	394.7
Boring Road	388.9
Danapur	388.9
Rajendra Nagar	379.7
Patliputra Colony	362.7
Total	8,442.7

FILTERS



4G NETWORK ANALYSIS

Analyze 4G network speed, signal strength, service providers, and technology performance

Service Provider

All

Technology

All

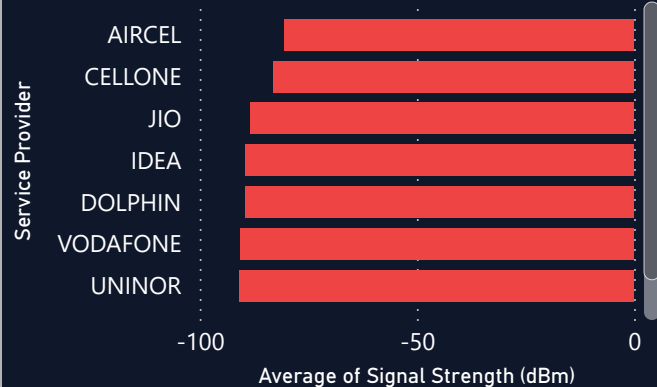
Test_type

All

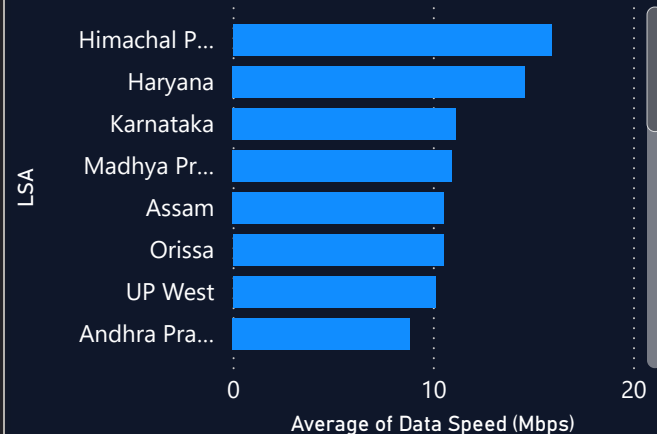
LSA

All

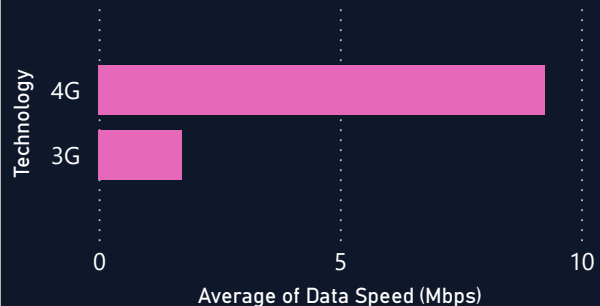
Avg Signal Strength by Service Provider



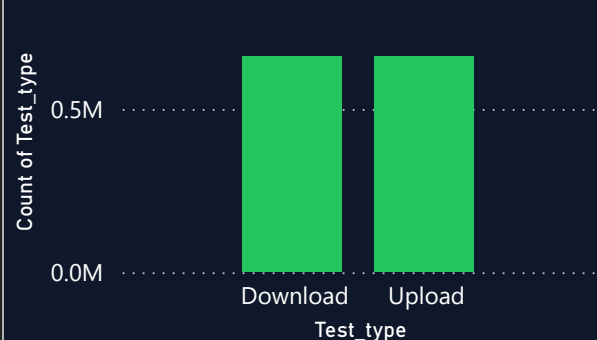
Average Data Speed by LSA



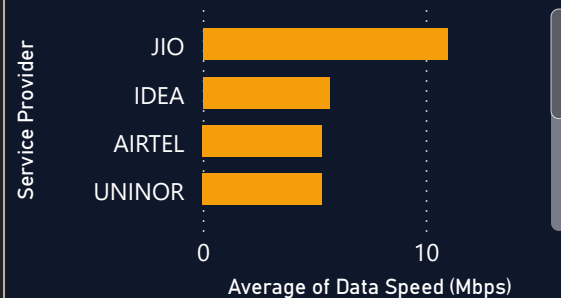
Avg Data Speed by Technology



Test Type Distribution



Avg Data Speed by Service Provider



Service Provider	Technology	LSA
AIRCEL	3G	Andhra Pradesh
AIRCEL	3G	Andhra Pradesh
AIRCEL	3G	Assam
AIRCEL	3G	Assam
AIRCEL	3G	Chennai
AIRCEL	3G	Chennai
AIRCEL	3G	Karnataka
AIRCEL	3G	Karnataka
AIRCEL	3G	Kolkata
AIRCEL	3G	Kolkata
AIRCEL	3G	Tamil Nadu
AIRCEL	3G	Tamil Nadu
AIRTEL	3G	Andhra Pradesh
AIRTEL	3G	Andhra Pradesh
AIRTEL	3G	Assam
AIRTEL	3G	Assam
AIRTEL	3G	Bihar
AIRTEL	3G	Bihar
AIRTEL	3G	Chennai
AIRTEL	3G	Chennai
AIRTEL	3G	Delhi
AIRTEL	3G	Delhi
AIRTEL	3G	Gujarat
AIRTEL	3G	Gujarat
Total		

-89.23

Avg Signal Strength

8

Total Service Providers

1.322972M

Total Technology

11.20M

Avg Data Speed





5G NETWORK ANALYSIS

Analyze download speed, upload speed, latency, signal strength, and network performance of the 5G network

5G

FILTERS

Timestamp

15-06-2024

28-05-2025

Carrier

All

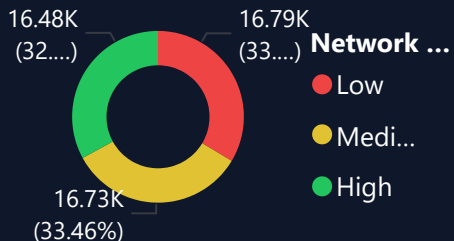
Location

All

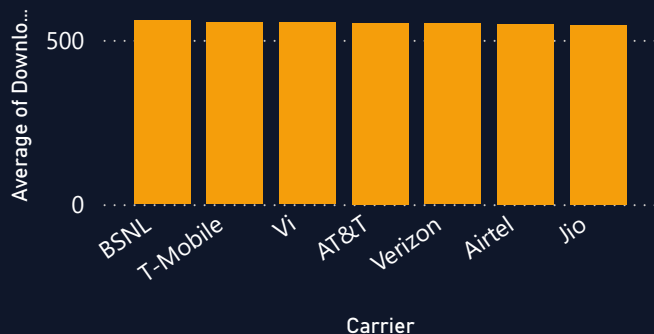
Device Model

All

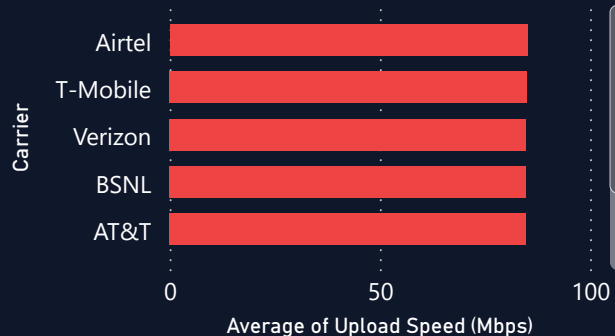
Network Congestion Level Distribution



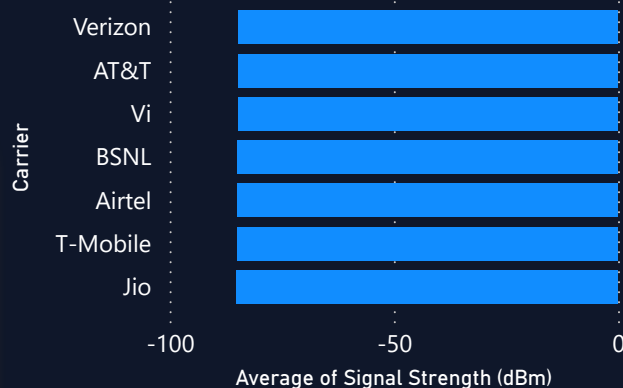
Download Speed by Carrier



Upload Speed by Carrier



Signal Strength by Carrier



Latency by Location



Carrier	Device Model	Average of Download Speed (Mbps)
Airtel	Galaxy S23	
Airtel	GT 7	
Airtel	iPhone 14	
Airtel	Nord 4	
Airtel	Pixel 7	
AT&T	Galaxy S23	
AT&T	GT 7	
AT&T	iPhone 14	
AT&T	Nord 4	
AT&T	Pixel 7	
BSNL	Galaxy S23	
BSNL	GT 7	
BSNL	iPhone 14	
BSNL	Nord 4	
BSNL	Pixel 7	
Jio	Galaxy S23	
Jio	GT 7	
Jio	iPhone 14	
Jio	Nord 4	
Total		

84.80

↑ Avg Upload Speed

-84.84

📶 Avg Signal Strength

551.18

↓ Avg Download Speed

10.50

🕒 Avg Latency

FILTERS

Network

All



6G NETWORK PREDICTION

Predicted performance metrics based on the analysis of 3G, 4G, and 5G network datasets



8.00

Predicted Latency

1.87K

Predicted Download...

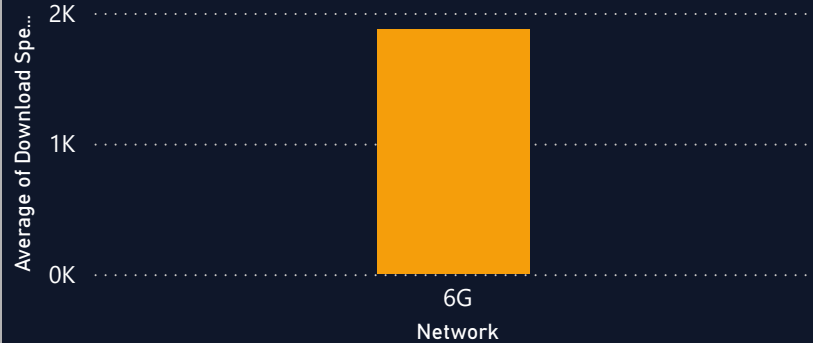
2.25K

Predicted Throughput

5

Predicted Range

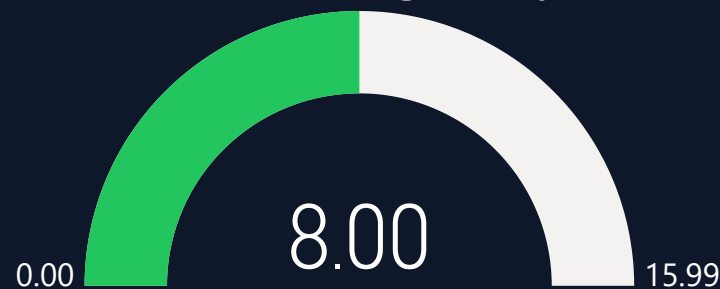
Predicted 6G Performance



Predicted Range



Predicted Avg Latency



Network	Sum of Download Speed(Mbps)	Sum of Latency(ms)	Sum of Upload Speed (Mbps)
6G	1,872.19	8.00	848.02
Total	1,872.19	8.00	848.02

The 6G network values are predicted using the average trends observed in the 3G, 4G, and 5G datasets. The prediction indicates significantly higher download speed, lower latency, improved signal quality, and wider coverage range compared to previous network generations.

Key influencers Top segments



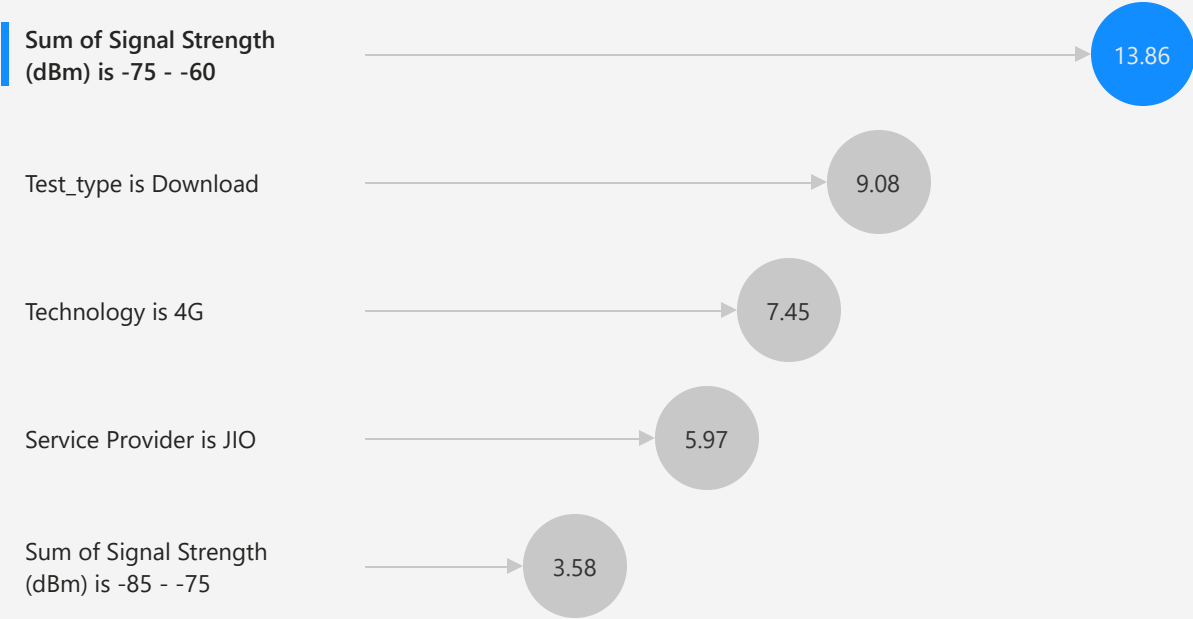
What influences Data Speed (Mbps) to

Increase

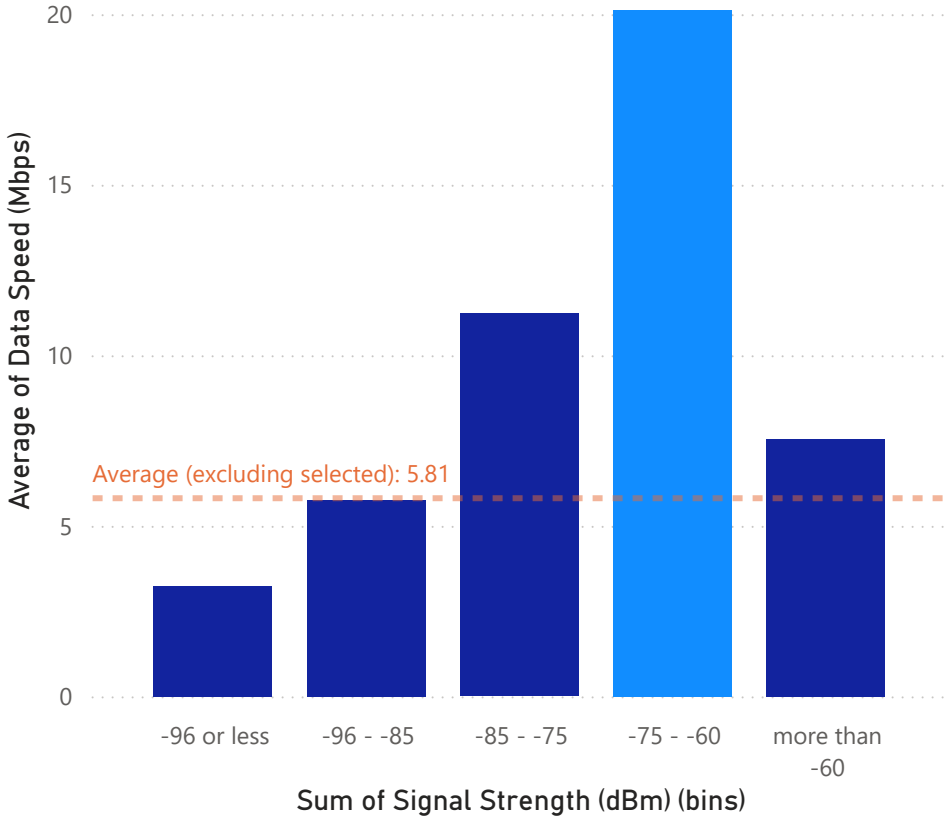
 ?

When...

...the average of Data Speed (Mbps) increases by



← Data Speed (Mbps) is more likely to increase when Sum of Signal Strength (dBm) is -75 - -60 than otherwise (on average).



☐ Only show values that are influencers



5G NETWORK PERFORMANCE ANALYTICS DASHBOARD

Real-time insights into network, performance, signal quality, connectivity, and user experience

This project analyzed the performance of 3G, 4G, and 5G networks and developed a predicted 6G network model based on the observed trends. The comparison shows continuous improvements in speed, latency, signal quality, and coverage across each generation. The predicted 6G network is expected to provide ultra-high speed, very low latency, wider coverage, and improved overall network performance, making it suitable for future communication technologies and intelligent applications.